Therefore, the modulus and principal argument of

$\sqrt{3}-i$ are 2 and $-\frac{\pi}{6}$.

In all the four cases, modulus are equal, but the arguments are depending on the quadrant in which the complex number lies.

Example 2.23

Represent the complex number (i) $-1-i$ (ii) $1+i\sqrt{3}$ in polar form.

Solution

(i)
$$\text{Let } -1-i = r(\cos\theta + i\sin\theta)$$

$$\text{We have } r = \sqrt{x^2+y^2} = \sqrt{1^2+1^2} = \sqrt{1+1} = \sqrt{2}$$

$$\alpha = \tan^{-1}\left|\frac{y}{x}\right| = \tan^{-1}1 = \frac{\pi}{4}.$$

Since the complex number $-1-i$ lies in the third quadrant, it has the principal value,

$$\theta = \alpha - \pi = \frac{\pi}{4} - \pi = -\frac{3\pi}{4}$$

$$\text{Therefore, } -1-i = \sqrt{2}\left(\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right)$$

$$= \sqrt{2}\left(\cos\frac{3\pi}{4} - i\sin\frac{3\pi}{4}\right).$$

$$-1-i = \sqrt{2}\left(\cos\left(\frac{3\pi}{4}+2k\pi\right) - i\sin\left(\frac{3\pi}{4}+2k\pi\right)\right),\ k\in\mathbb{Z}.$$

Note

Depending upon the various values of k, we get various alternative polar forms.

(ii) $1+i\sqrt{3}$

$$r = |z| = \sqrt{1^2+\left(\sqrt{3}\right)^2} = 2$$

$$\theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{3}$$

$$\text{Hence } \arg(z) = \frac{\pi}{3}.$$

Therefore, the polar form of $1+i\sqrt{3}$ can be written as

$$1+i\sqrt{3} = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

$$= 2\left(\cos\left(\frac{\pi}{3}+2k\pi\right) + i\sin\left(\frac{\pi}{3}+2k\pi\right)\right),\ k\in\mathbb{Z}.$$

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